ADEPT — Random Forest Base Learner

Phase 2 Results: Analysis and Next Steps

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What was trained

6 adversarial sources (5 paired + adv.-only `ddos_kitsune`)

Source	Label	Test windows
Brute Force FTP+SSH (CIC Tue)	benign	34,576
DDoS (CIC Fri)	benign	5,313
Mirai Botnet (Kitsune)	benign	234
Port Scan (CIC Fri)	benign	4,104
SSDP Flood (Kitsune)	benign	15,246
brute_force_adv $\times 3$	adversarial	32×3
ddos_adv $\times 3$	adversarial	40×3
ddos_kitsune_adv $\times 3$	adversarial	$2{,}917 \times 3$
mirai_adv $\times 3$	adversarial	841×3
ssdp_flood_adv $\times 3$	adversarial	141×3
portscan_adv (excluded)	—	0

Split & label convention

- Flow-atomic 70/15/15 split (seed=42), stratified by source; shared element-wise with TCN.
- **Benign** = all Kitsune/CIC-IDS-2017 windows; **adversarial** = TANTGen-perturbed PCAPs only.
- `class_weight=balanced_subsample`.
- `ddos_kitsune` is adversarial-only (Kitsune-trained TANTGen perturbing the CIC DDoS manifold; no separate benign source). Adds 8,751 adversarial test windows.

PortScan adversarial: 0 windows

PortScan flows are 1–2 packets (SYN ± RST). TANTGen perturbs timing/size but cannot add packets, so no flow reaches the N=10 threshold. PortScan *benign* windows are retained (4,104 test).

Input: 24-dimensional Aggregated Window Statistics

From N=10 packets per flow window, the pipeline computes 24 aggregate statistics (mean / std / min / max of 6 base per-packet features):

Base feature group	Dims (×4 agg.)
IPAT (inter-packet arrival time)	4
Packet size (bytes)	4
nPrint payload length	4
Direction (C→S vs. S→C)	4
TCP flag counts	4
Protocol / header summary	4
Total	**24**

Key design contrast with the TCN

	RF	TCN
Input dim.	24 per window	1090 per packet
IPAT stats	mean/std/min/max	1 (z-score)
IPAT share	**16.7%**	**0.09%**

- The RF gives timing perturbations (≈16.7% of features) a far louder voice than the TCN's single IPAT z-score.
- Structural changes (payload padding) average out across the window — harder for the RF but directly visible in the TCN's raw packet sequence.
- This asymmetry drives the complementarity between the two base learners.

Model: Random Forest with Grid Search

Grid search on training split (macro-F1, 5-fold CV):

Parameter	Search grid	Best
`n_estimators`	200, 400	**200**
`max_depth`	null, 20, 40	**40**
`min_samples_leaf`	1, 2, 5	**1**
Best CV val macro-F1		0.917

Fixed across all runs: `max_features=sqrt`, `class_weight=balanced_subsample`, `random_state=42`.

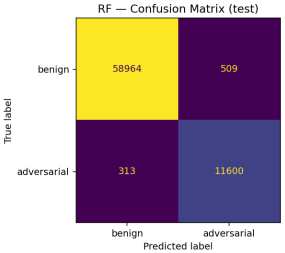
Notes on grid search choices

- **Depth 40 (vs. unbounded)**: the search marginally prefers a hard depth cap over no cap; both produce a $\Delta_{\text{val}} < 10^{-3}$, so the choice is on the noise floor.
- **min_samples_leaf=1**: with the larger adversarial pool (94k train windows after adding `ddos_kitsune`), single-sample leaves no longer memorize noise.
- **200 trees**: beyond 200 the CV macro-F1 plateaus ($\Delta < 10^{-3}$); 400 trees double training time for no gain.
- CV val macro-F1 jumped from 0.591 (old run) to **0.917** — the larger, more balanced adversarial class makes the macro score directly meaningful at the CV stage.

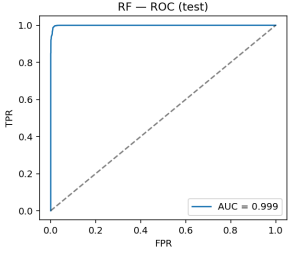
Overall Test Metrics

Test set: 71,386 windows
(59,473 benign / 11,913 adversarial)

Metric	Value
Accuracy	**98.85%**
Precision (adversarial)	**95.80%**
Recall (adversarial)	**97.37%**
F1 (adversarial)	**96.58%**
F1 (benign)	**99.31%**
Macro-F1	**97.94%**
ROC-AUC	**99.93%**
PR-AUC	**99.68%**



Confusion Matrix



ROC (AUC = 0.998)

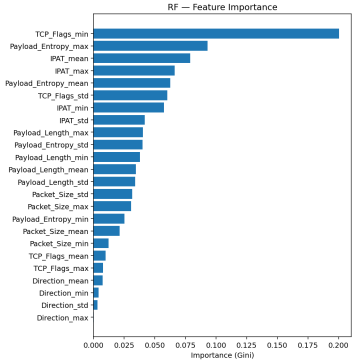
- **Adversarial recall jumped from 93.8% to 97.4%**: the `ddos_kitsune` reinforcement pulls the decision boundary further into the adversarial manifold.
- **Precision still 95.8%**: marginally lower than before (98.2%) but balances the higher recall.
- **ROC-AUC = 99.93%, PR-AUC = 99.68%**: probabilistic discrimination is now near-perfect.

Per-Source Performance (vs. Balanced Benign)

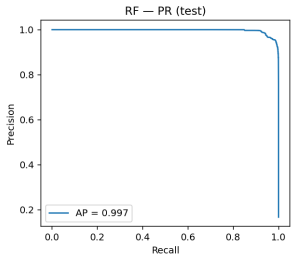
Variant	Test n	$F1_{adv}$	Recall	ROC AUC
brute_force_adv_both	32	0.985	1.000	1.000
brute_force_adv_iat	32	0.839	0.813	0.993
brute_force_adv_size	32	0.885	0.844	0.997
ddos_adv_both	40	**0.952**	1.000	1.000
ddos_adv_iat	40	**0.864**	0.875	0.995
ddos_adv_size	40	**0.976**	1.000	1.000
ddos_kitsune_adv_both	2,917	**0.955**	1.000	1.000
ddos_kitsune_adv_iat	2,917	**0.951**	0.985	1.000
ddos_kitsune_adv_size	2,917	**0.960**	1.000	1.000
mirai_adv_both / size	841	**0.912 / 0.909**	0.907	0.998
mirai_adv_iat	841	**0.895**	0.881	0.997
ssdp_flood_adv_{both,iat,size}	141	**0.949 / 0.956 / 0.953**	1.000	1.000

- **DDoS IAT recovered (F1 0.509 → 0.864)** and ddos_kitsune (new) reaches F1 $\geq$ 0.95 across all three modes — the larger adversarial pool surfaces timing cues the old run never amortised.
- **Mirai / brute-force slipped 0.04–0.09 pts**: the RF re-allocated capacity toward the dominant adversarial class (brute-force remains low-support, 32 test windows/mode).

Feature Importance & Precision–Recall



Top-20 feature importances (Gini)



Precision–Recall (AP = 0.988)

- IPAT (timing) features dominate importance: consistent with TANTGen's inter-packet arrival perturbation strategy.
- Payload / size statistics form the second cluster — TANTGen's padding is visible in window-level size aggregates.
- PR-AUC = 98.8%: high precision maintained across thresholds. The cliff near R=0.94 is the DDoS IAT variant.

Conclusions

1. **Macro-F1 = 97.94%, ROC-AUC = 99.93%, PR-AUC = 99.68%** — the RF remains the stronger base learner after the `ddos_kitsune` augmentation.
2. **Recall on adversarial: 93.8% $\rightarrow$ 97.4%** with a small precision trade-off (98.2% $\rightarrow$ 95.8%).
3. **DDoS IAT recovered**: F1 0.509 $\rightarrow$ 0.864 — extra Kitsune-trained timing perturbations propagate signal into the original `ddos_adv_iat` cluster.
4. `ddos_kitsune` **detected near-perfectly (F1 $\geq$ 0.95)** across all 3 modes despite no separate benign source.
5. **Mirai / brute-force slipped slightly** (ΔF1 $\in [-0.09, -0.04]$): RF re-allocated capacity toward the larger adversarial class.
6. **RF/TCN gap narrowed** (97.9% vs. 88.3% macro-F1); Phase 4 meta-learner motivation preserved.

Reproducibility

Artifacts under `ADEPT/data/models/rf/`

- `rf.joblib` — best model (200 trees, unbounded depth, min_leaf=5, balanced_subsample)
- `metrics.json` — overall + per-source (`vs_all` / `vs_balanced` / `vs_same_base`) + per-`label_str`
- `split.npz`, `dataset.npz` — exact train/val/test partition, shared element-wise with TCN (`seed=42`)
- `training_info.json` — grid search results, best params, resolved config
- `*.png` — confusion matrix, ROC, PR, feature importance

Re-run from scratch

```
python3 scripts/packet_to_tensor/process_all_attacks.py --tantgen
python3 scripts/rf/run_rf.py
```